

5G RAN

MRO Feature Parameter Description

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Huawei Technologies Co., Ltd.

Address: Huawei Industrial Base
Bantian, Longgang
Shenzhen 518129
People's Republic of China

Website: <https://www.huawei.com>

Email: support@huawei.com

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1 Change History

This chapter describes changes not included in the "Parameters", "Counters", "Glossary", and "Reference Documents" chapters. These changes include:

- Technical changes
Changes in functions and their corresponding parameters
- Editorial changes
Improvements or revisions to the documentation

1.1 5G RAN10.1 Draft A (2025-12-31)

This issue introduces the following changes to 5G RAN9.1 01 (2025-03-10).

Technical Changes

Change Description	Parameter Change	RAT	Base Station Model
Added an impact relationship between cell-level MRO and RF module fast deep dormancy. For details, see 4.3.2.3 Function Impacts .	None	Low-frequency TDD	3900 and 5900 series base stations (macro base stations)
Added an impact relationship between UE-level MRO and RF module fast deep dormancy. For details, see 5.3.2.3 Function Impacts .	None	Low-frequency TDD	3900 and 5900 series base stations (macro base stations)

Editorial Changes
None

2 About This Document

2.1 General Statements

Purpose

Feature Parameter Description documents are intended to acquaint readers with:

- The technical principles of features and their related parameters
- The scenarios where these features are used, the benefits they provide, and the impact they have on networks and functions
- Requirements that must be met before feature activation
- Parameter configuration required for feature activation, verification of feature activation, and monitoring of feature performance

NOTE

This document only provides guidance for feature activation. Feature deployment and feature gains depend on the specifics of the network scenario where the feature is deployed. To achieve optimal gains, contact Huawei professional service engineers.

Functions mentioned in this document work properly only when enabled in the specified applicable scenarios (such as RAT and networking). If a function not mentioned in this document is enabled or a function is enabled in a scenario not specified as applicable, exceptions or other impacts may occur.

Software Interfaces

Any parameters, alarms, counters, or managed objects (MOs) described in Feature Parameter Description documents apply only to the corresponding software release. For future software releases, refer to the corresponding updated product documentation.

2.2 Features in This Document

This document describes the following features.

Feature ID	Feature Name	Chapter/Section
FOFD-070320	Mobility Robustness Optimization	4 Cell-Level MRO 5 UE-Level MRO

2.3 Differences

Differences Between NSA and SA

Function Name	Difference	Chapter/Section
Cell-level MRO	Supported only in SA networking	4 Cell-Level MRO
UE-level MRO	Supported only in SA networking	5 UE-Level MRO

Differences Between High Frequency Bands and Low Frequency Bands

This document refers to frequency bands belonging to FR1 (410–7125 MHz) as low frequency bands, and those belonging to FR2 (24250–71000 MHz) as high frequency bands. For details about FR1 and FR2, see section 5.1 "General" in 3GPP TS 38.104 V17.8.0.

Function Name	Difference	Chapter/Section
Cell-level MRO	Supported only in low frequency bands	4 Cell-Level MRO
UE-level MRO	Supported only in low frequency bands	5 UE-Level MRO

3 Overview

As mobile telecommunications technologies advance, wireless networks continue to grow and incorporate multiple radio access technologies (RATs), resulting in complicated network maintenance. In addition, improper settings of handover or radio frequency (RF) parameters may cause handover exceptions in the network, affecting user experience.

To simplify network maintenance, an NR system must support the self-organizing network (SON). Mobility robustness optimization (MRO), as one of the key technologies of SON, **automatically identifies abnormal handover scenarios and optimizes handover parameter settings. This reduces abnormal handovers, lowers the network planning, optimization, and O&M costs, and improves network optimization efficiency.**

gNodeBs support only intra-RAT intra-frequency MRO, which is classified into cell-level MRO and UE-level MRO.

- Cell-level MRO: The gNodeB automatically identifies cell-level abnormal handover scenarios, such as premature handovers, delayed handovers, handovers to a wrong cell, and coverage-induced abnormal handovers, collects statistics on abnormal handovers, and automatically optimizes cell-level handover parameters based on the statistics. Abnormal coverage is usually caused by inappropriate RF parameter settings. Therefore, the gNodeB does not optimize coverage-induced abnormal handovers.
- UE-level MRO: The gNodeB automatically identifies UE-level abnormal handover scenarios, such as ping-pong handovers, collects statistics on abnormal handovers, and automatically optimizes UE-level handover parameters based on the statistics.

4 Cell-Level MRO

4.1 Principles

4.1.1 Scenario Identification

The gNodeB can automatically identify abnormal handover scenarios, including premature handovers, delayed handovers, handovers to a wrong cell, and coverage-induced abnormal handovers.

This function is controlled by the **MRO_SCENARIO_IDENT_SW** option of the **gNBMr.ScenarioidentifySwitch** parameter. If this option is selected, the gNodeB can identify the preceding abnormal handover scenarios and measure the number of abnormal handovers using related counters.

4.1.1.1 Premature Handovers

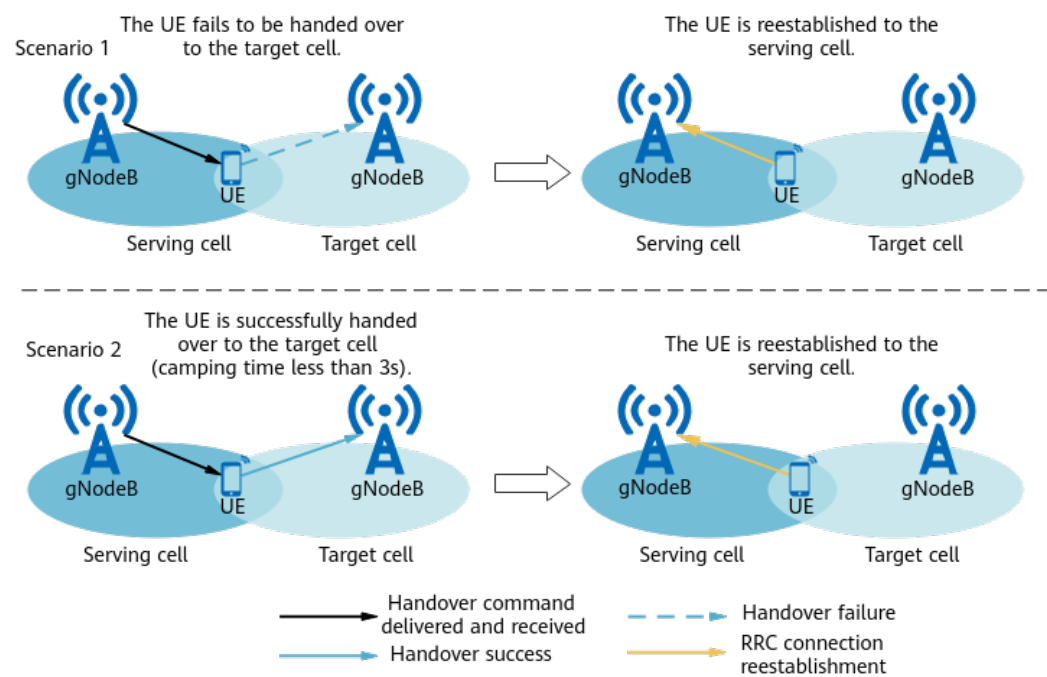
A premature handover occurs when the target cell is too likely to meet the handover conditions or the signal quality is unstable. As a result, a radio link failure (RLF) occurs and the UE reestablishes its radio resource control (RRC) connection to the original serving cell. After automatically identifying a premature handover, the gNodeB counts the number of premature handovers using the **N.HO.IntraRAT.HoTooEarly** and **N.HO.NCell.HoTooEarly** counters.

Table 4-1 and **Figure 4-1** show scenarios of premature handovers. gNodeBs identify premature handovers based on the results of handovers to the target cell, duration of camping in the target cell, and RRC connection reestablishment results, as listed in **Table 4-1**.

Table 4-1 Scenarios of premature handovers

Scena rio	Result of the Handover to the Target Cell	Duration of Camping in the Target Cell	RRC Connec tio n Reestablis hment Result	Exception Cause
Scenar io 1	Failed	No camping	Reestablis hed to the serving cell	The target cell is too likely to meet the handover conditions or the signal quality is unstable.
Scenar io 2	Succeeded	Camping time less than 3s	Reestablis hed to the serving cell	

Figure 4-1 Scenarios of premature handovers



NOTE

MRO supports both intra-base-station and inter-base-station scenarios. The figures in this document use inter-base-station scenarios as examples.

If a UE repeatedly experiences abnormal handovers such as premature handovers in scenario 1, the gNodeB cannot correctly measure the number of premature handovers. As a result, the accuracy of the handover parameter optimization cannot be ensured.

4.1.1.2 Delayed Handovers

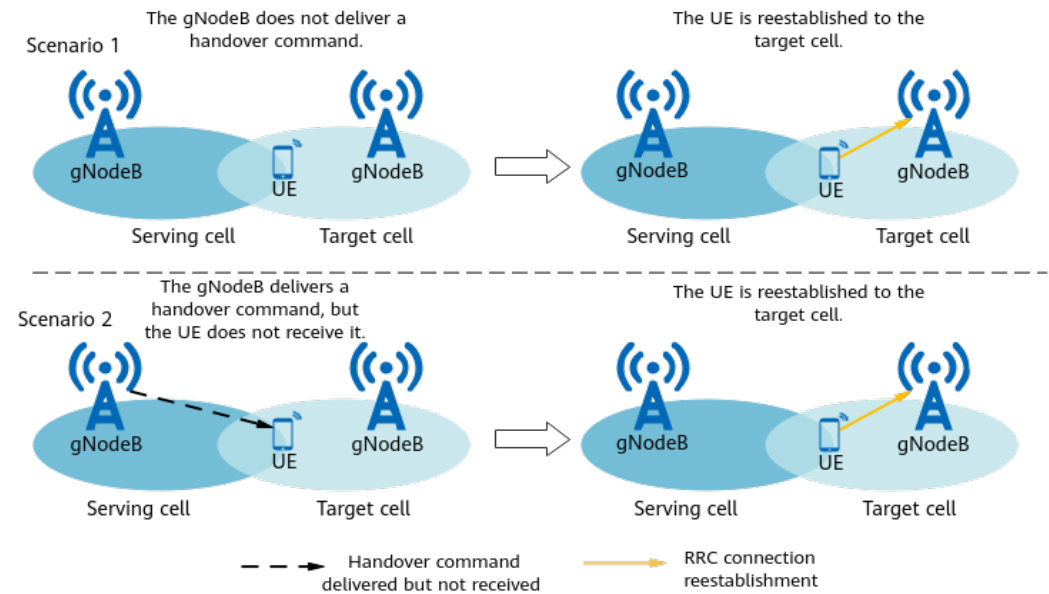
A delayed handover occurs when the target cell is too difficult to meet the handover conditions. The handover procedure fails to be triggered or the UE does not receive a handover command after the UE moves out of the coverage of the serving cell. As a result, the UE experiences an RLF in the serving cell and reestablishes its RRC connection to the target cell. After automatically identifying a delayed handover, the gNodeB counts the number of delayed handovers using the **N.HO.IntraRAT.HoTooLate** and **N.HO.NCell.HoTooLate** counters.

Table 4-2 and **Figure 4-2** show scenarios of delayed handovers. gNodeBs identify delayed handovers based on the results of handovers to the target cell, duration of camping in the target cell, and RRC connection reestablishment results, as listed in **Table 4-2**.

Table 4-2 Scenarios of delayed handovers

Scenario	Result of the Handover to the Target Cell	Duration of Camping in the Target Cell	RRC Connection Reestablishment Result	Exception Cause
Scenario 1	No handover (The gNodeB does not deliver a handover command.)	No camping	Reestablished to the target cell	The target cell is too difficult to meet the handover conditions. When the UE moves out of the coverage of the serving cell, it still cannot meet the measurement reporting conditions. As a result, the handover procedure cannot be triggered.
Scenario 2	No handover (The gNodeB delivers a handover command, but the UE does not receive it.)	No camping	Reestablished to the target cell	The target cell is too difficult to meet the handover conditions. When the UE moves out of the coverage of the serving cell, the signal quality of the serving cell is poor and the UE fails to receive a handover command.

Figure 4-2 Scenarios of delayed handovers



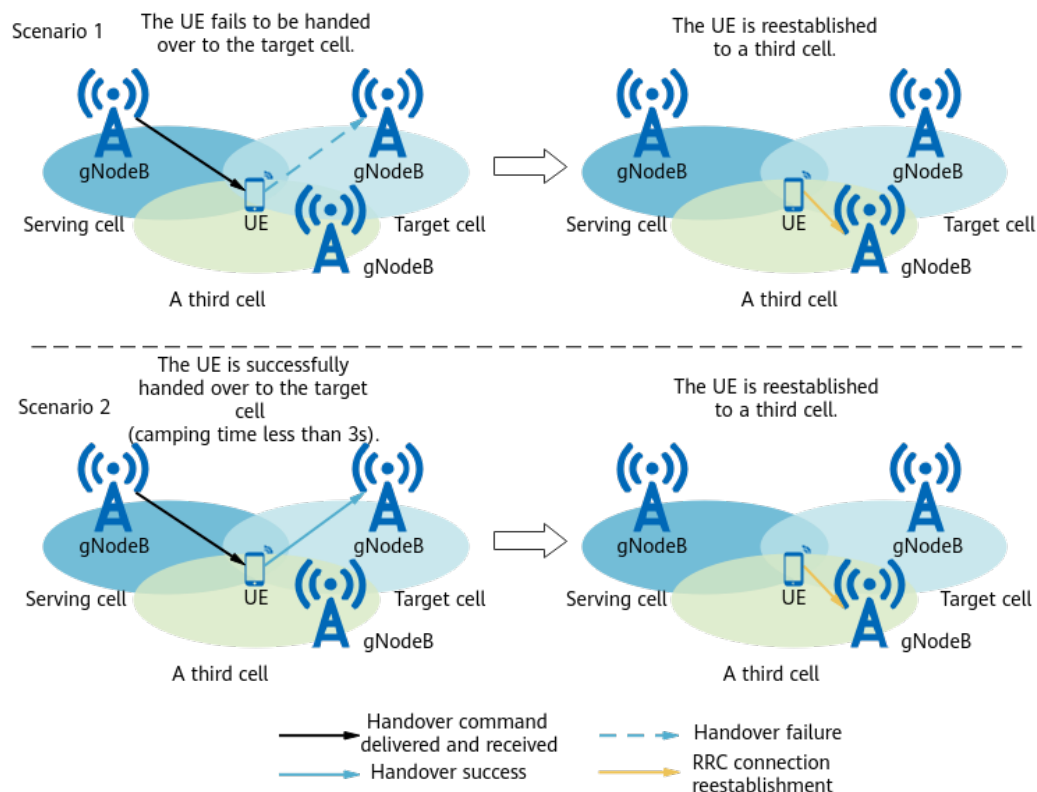
4.1.1.3 Handovers to a Wrong Cell

A handover to a wrong cell occurs when the target cell is too likely to meet the handover conditions and its signal quality is unstable, but a third cell is too difficult to meet the handover conditions and its signal quality is good. In this case, the UE is reestablished to the third cell. After automatically identifying a handover to a wrong cell, the gNodeB counts the number of handovers to a wrong cell using the **N.HO.IntraRAT.HoToWrgCell** and **N.HO.NCell.HoToWrgCell** counters.

Table 4-3 and **Figure 4-3** show the scenarios of handovers to a wrong cell. gNodeBs identify handovers to a wrong cell based on the results of handovers to the target cell, duration of camping in the target cell, and RRC connection reestablishment results, as listed in **Table 4-3**.

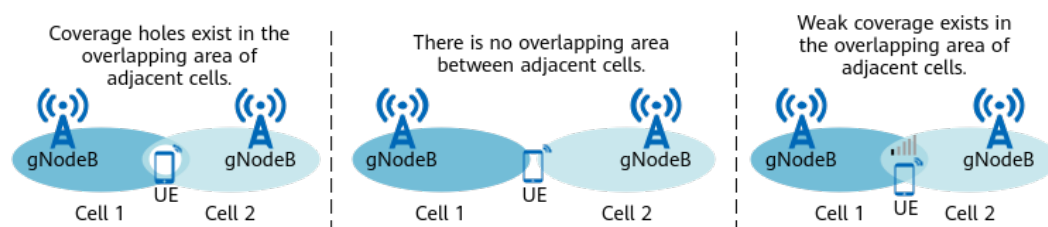
Table 4-3 Handovers to a wrong cell

Scenar io	Result of the Handover to the Target Cell	Duration of Camping in the Target Cell	RRC Connec tio n Reestablis hment Result	Exception Cause
Scenar io 1	Failed	No camping	Reestablis hed to a third cell	The target cell is too likely to meet the handover conditions and its signal quality is unstable, while the third cell is too difficult to meet the handover conditions and its signal quality is good.
Scenar io 2	Succeeded	Camping time less than 3s	Reestablis hed to a third cell	

Figure 4-3 Handovers to a wrong cell


4.1.1.4 Coverage-induced Abnormal Handovers

If an RLF occurs during a handover, the UE cannot initiate an RRC connection reestablishment due to abnormal coverage. After the UE accesses the cell again, the RSRP values of all intra-frequency neighboring cells and the serving cell contained in the RLF Report message are less than the value of **NRCellMro.AbnCoverageRsrpThld**, and the RLF Report message does not contain the *reestablishmentCellId-r16* IE. In this case, the gNodeB considers that the RLF during the handover is caused by abnormal coverage. Abnormal coverage is usually caused by improper RF parameter settings, as shown in Figure 4-4. After automatically identifying coverage-induced abnormal handovers, the gNodeB counts the number of RLFs caused by abnormal coverage using the **N.RLF.CoverageAbnormal** counter.

Figure 4-4 Scenarios of abnormal coverage


4.1.2 Optimization

After identifying premature handovers, delayed handovers, or handovers to a wrong cell, the gNodeB triggers handover parameter optimization if the corresponding conditions are met. Then, the gNodeB implements the optimization advice directly (or as instructed by the MAE-Access) based on the optimization mode. Currently, only the handover parameter cell individual offset (CIO) can be optimized.

NOTE

Abnormal coverage is usually caused by inappropriate RF parameter settings. Therefore, the gNodeB does not optimize coverage-induced abnormal handovers.

4.1.2.1 Optimization Triggering

Within an MRO optimization period specified by **gNBMrro.OptPeriod**:

- If the number of intra-frequency handovers is greater than or equal to the value of **gNBMrro.StatNumThld**, handover parameter optimization is triggered.
- If the number of intra-frequency handovers is less than the value of **gNBMrro.StatNumThld**, the number of handovers is accumulated in the next MRO optimization period until it is greater than or equal to the value of **gNBMrro.StatNumThld**. Meanwhile, handover parameter optimization is triggered. Therefore, the optimization period is the sum of all MRO optimization periods during which handover parameter optimization is triggered.

The number of intra-frequency handovers includes the number of executed intra-frequency outgoing handovers and the number of delayed handovers (scenario 1).
Number of executed intra-frequency outgoing handovers =
N.HO.IntraFreq.Xn.IntergNB.ExecAttOut + **N.HO.IntraFreq.IntragNB.ExecAttOut**
+ **N.HO.IntraFreq.Ng.IntergNB.ExecAttOut**.

4.1.2.2 Optimization Execution

Optimization Mode Configuration

The MRO optimization mode is specified by the **NR_INTRA_RAT_MRO_CTRL_MODE_SW** option of the **NRCellMrro.ModeSwitch** parameter.

- When this option is deselected, the MRO optimization mode is free mode.
After an MRO optimization period ends, the gNodeB determines the handover parameter to be optimized based on abnormal handover statistics and implements the parameter optimization advice.
- When this option is selected, the MRO optimization mode is controlled mode.
After an MRO optimization period ends, the gNodeB determines the handover parameters to be optimized based on abnormal handover statistics, and reports the optimization advice to the MAE-Access. The MAE-Access then delivers the parameter optimization advice confirmed by the maintenance personnel to the gNodeB for taking effect. In addition, the maintenance

personnel can manually modify parameter optimization advice on the MAE-Access.

In controlled mode, the MAE-Access generates MRO optimization advice, which contains statistics on MRO-related abnormal handovers, as listed in [Table 4-4](#).

Table 4-4 Statistics on MRO-related abnormal handovers

Item	Description
Handover Execution Attempts Count	Indicates the number of outgoing intra-frequency handover execution attempts.
Successful Handover Count	Indicates the number of successful intra-frequency handovers.
Early Handover Count	Indicates the number of premature intra-frequency handovers.
Late Handover Count without HO Execution	Indicates the number of delayed intra-frequency handovers without handover command delivery.
Late Handover Count with HO Execution	Indicates the number of delayed intra-frequency handovers with handover command delivery.
Wrong Handover Count	Indicates the number of intra-frequency handovers to a wrong cell.

NOTE

- The items in the [Table 4-4](#) are measured and their values are accumulated within all MRO optimization periods during which handover parameter optimization is triggered.
- These items are measured by MRO based on handover procedures and RLFs, and may differ from those described in performance counter reference.
- If the gNodeB automatically determines not to perform optimization based on the statistics about abnormal handovers, it does not generate or report optimization advice to the MAE-Access.

Parameter Optimization

After handover parameter optimization is triggered, the gNodeB automatically optimizes the *NRCellRelation.CellIndividualOffset* parameter based on the identification of abnormal intra-frequency handovers and statistics when the optimization period ends. Parameter optimization for cell-level intra-frequency MRO is controlled by the **INTRA_FREQ_MRO_SW** option of the **NRCellMro.MroSwitch** parameter.

When this option is selected, if the number of intra-frequency handovers is greater than or equal to the value of **gNBmro.StatNumThld** and the proportion of RLF-induced abnormal handovers from the serving cell to the target cell is greater than the value of **gNBmro.AbnHoRatioThld** within an optimization period, then the gNodeB determines whether premature handovers, delayed handovers, or

handovers to a wrong cell occur. In addition, when an optimization period ends, the gNodeB plans to optimize the CIO only in the following scenarios:

- If the proportion of the sum of the number of premature handovers and the number of handovers to a wrong cell to the number of RLF-induced abnormal handovers is greater than the value of **gNBmro.TooEarlyHoWrongHoRatioThld**, the gNodeB plans to decrease the CIO of the intra-frequency neighboring cell by one step. That is, the value of **NRCellRelation.CellIndividualOffset** is decreased by one enumerated value, for example, from **DB_10** to **DB_12** or from **DB5** to **DB4**.
- If the proportion of delayed handovers to the number of RLF-induced abnormal handovers is greater than the value of **gNBmro.TooLateHoRatioThld**, the gNodeB plans to increase the CIO of the intra-frequency neighboring cell by one step. That is, the value of **NRCellRelation.CellIndividualOffset** is increased by one enumerated value, for example, from **DB_10** to **DB_8** or from **DB5** to **DB6**.

All the following items are measured in intra-frequency handover scenarios:

- Number of intra-frequency handovers = Number of outgoing handover executions + Number of delayed handovers (scenario 1)
- Number of outgoing handover executions = **N.HO.IntraFreq.Xn.IntergNB.ExecAttOut** + **N.HO.IntraFreq.IntragNB.ExecAttOut** + **N.HO.IntraFreq.Ng.IntergNB.ExecAttOut**
- Proportion of RLF-induced abnormal handovers = (Number of premature handovers + Number of delayed handovers + Number of handovers to a wrong cell)/Number of intra-frequency handovers
- Number of RLF-induced abnormal handovers = Number of premature handovers + Number of delayed handovers + Number of handovers to a wrong cell
- Number of premature handovers (indicated by the **N.HO.NCell.HoTooEarly** counter)
- Number of delayed handovers (indicated by the **N.HO.NCell.HoTooLate** counter)
- Number of handovers to a wrong cell (indicated by the **N.HO.NCell.HoToWrgCell** counter)

The gNodeB considers the valid CIO value range [CioAdjLowerLimit, CioAdjUpperLimit] during CIO optimization.

- $\text{CioAdjLowerLimit} = \text{NRCellIntraFHoMeaGrp.IntraFreqHoA3Offset} + \text{NRCellIntraFHoMeaGrp.IntraFreqHoA3Hyst} - 5$
- $\text{CioAdjUpperLimit} = \text{NRCellIntraFHoMeaGrp.IntraFreqHoA3Offset} + \text{NRCellIntraFHoMeaGrp.IntraFreqHoA3Hyst} - 2$

When an optimization period ends, the gNodeB checks whether each NRCellIntraFHoMeaGrp instance in the cell meets any of the following conditions. If yes, the gNodeB performs CIO optimization as planned.

- The CIO value after optimization is within the valid CIO value range.
- The CIO value after optimization is greater than the value of CioAdjUpperLimit, and the planned optimization measure is to decrease the CIO value by one step.

- The CIO value after optimization is smaller than the value of **CioAdjLowerLimit**, and the planned optimization measure is to increase the CIO value by one step.

If the value of **NRCellRelation.CellIndividualOffset** is changed within an MRO optimization period, the gNodeB does not perform MRO and clears the number of handovers after this period ends, but starts to perform MRO only after the next optimization period ends.

4.2 Network Analysis

4.2.1 Benefits

Cell-level intra-frequency MRO, after being enabled, can reduce the number of abnormal handovers (**N.HO.NCell.HoTooEarly**, **N.HO.NCell.HoTooLate**, and **N.HO.NCell.HoToWrgCell**), lower network planning, optimization, and O&M costs, and improve network optimization efficiency.

NOTE

Gains of cell-level intra-frequency MRO are subject to stable MRO traffic models. To better know the gains, you are advised to enable cell-level intra-frequency MRO when the handover area becomes stable. A stable handover area means that the traffic models remain unchanged for more than three consecutive traffic periods. You are advised to set the traffic period to one day. Traffic models refer to the average number of UEs (**N.User.RRCCConn.Avg**) and the number of handovers (**N.HO.IntraFreq.Xn.IntergNB.ExecAttOut**, **N.HO.IntraFreq.Ng.IntergNB.ExecAttOut**, and **N.HO.IntraFreq.IntragNB.ExecAttOut**).

You are advised to enable cell-level intra-frequency MRO in the following scenarios:

- Network deployment
Generally, RF engineering optimization is performed multiple times before a network is put into commercial use. It is recommended that cell-level intra-frequency MRO be enabled following RF engineering optimization.
- Large-scale network capacity expansion
Network capacity expansion affects the coverage of sites that do not require capacity expansion. Further, the neighbor relationships and handover areas of sites that do not require capacity expansion are altered, and new handover areas are introduced. Consequently, the original handover parameter settings become inappropriate. As such, you are advised to disable cell-level intra-frequency MRO on sites that do not require capacity expansion in areas demanding large-scale capacity expansion, set the CIO to 0 (**NRCellRelation.CellIndividualOffset** to 0), and then enable cell-level intra-frequency MRO on these sites after the network capacity is expanded and RF engineering optimization is complete.
- Sparse network capacity expansion (adding a few sites or cells) or scenarios where RF engineering optimization is performed only for some cells
Similar to large-scale capacity expansion scenarios, these scenarios affect neighbor relationships and handover areas of existing cells or sites that do not require capacity expansion. Therefore, you are advised to disable cell-level

intra-frequency MRO on sites that are affected by capacity expansion or RF engineering optimization, set the CIO to 0, and then enable cell-level intra-frequency MRO on these sites after the sparse network capacity expansion or RF engineering optimization is complete.

In addition, in areas with limited coverage and strong interference, you are advised to perform RF engineering optimization to enhance coverage or eliminate interference before enabling cell-level intra-frequency MRO. Cell-level intra-frequency MRO is not recommended in a heterogeneous network (HetNet).

4.2.2 Impacts

The impacts between this function and other functions are described in [4.3.2.3 Function Impacts](#).

Cell-level intra-frequency MRO aims to control the proportion of RLF-induced abnormal handovers below a specified threshold. Measures against premature handovers and handovers to a wrong cell are contradictory to those against delayed handovers. As such, measures against premature handovers or handovers to a wrong cell may increase the probability of delayed handovers and decrease the number of outgoing handovers, which may decrease the handover success rate but does not cause more handover failures. On the contrary, measures against delayed handovers may increase the probability of premature handovers and handovers to a wrong cell, as well as the number of outgoing handovers.

- Cell-level intra-frequency MRO against premature handovers and handovers to a wrong cell: If the CIO is adjusted to an excessively small value, handovers may not be triggered even though the signal strength of the neighboring cell is much higher than that of the serving cell. In this case, channel quality of some UEs in the serving cell becomes worse and the throughput decreases.
- Cell-level intra-frequency MRO against delayed handovers: If the CIO is adjusted to an excessively large value, handovers may be triggered once the signal strength of the neighboring cell is lower than that of the serving cell. In this case, channel quality of some UEs in the target cell becomes worse. As a result, the throughput decreases.

where

- Counters related to the number of outgoing handovers:
N.HO.IntraFreq.Xn.IntergNB.ExecAttOut,
N.HO.IntraFreq.Ng.IntergNB.ExecAttOut, and
N.HO.IntraFreq.IntragNB.ExecAttOut
- Throughput-related KPIs: **User Downlink Average Throughput (DU)**, **User Uplink Average Throughput (DU)**, **Cell Downlink Average Throughput (DU)**, and **Cell Uplink Average Throughput (DU)**

After the **MRO_SCENARIO_IDENT_SW** option of the **gNBMrro.ScenarioidentifySwitch** parameter is selected, the number of messages over the air interface and Xn interface increases. When control-plane messages over the air interface or Xn interface are congested, messages are discarded. As a result, the corresponding functions become abnormal.

4.3 Requirements

4.3.1 Licenses

Feature ID	Feature Name	Model	Sales Unit
FOFD-070320	Mobility Robustness Optimization	NR0S00MRPT00	per Cell

NOTE

The insufficiency of certain feature licenses affects cell activation. For details, see the **Is Cell Activation Affected by License Insufficiency** column in *License Control Item Lists*.

4.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

4.3.2.1 Prerequisite Functions

None

4.3.2.2 Mutually Exclusive Functions

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Adaptive handover in high-speed cells	NRCellMobilityConfig.HsCellAdaptHoOffsetLevel set to a value other than 0	<i>High Speed Mobility</i>	Adaptive handover in high-speed cells is mutually exclusive with cell-level intra-frequency MRO.
Low-frequency TDD	Smart SSB beam	SMART_SSB_BEAM_SW option of the NRDUCellAlgoSwitch.PowerSavingSwitch parameter	None	Smart SSB beam is mutually exclusive with cell-level intra-frequency MRO.

4.3.2.3 Function Impacts

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Intra-base-station UL CoMP	INTRA_GNB_UL_COMP_SW option of the NRDUCellAIgoSwitch.CompSwitch parameter	<i>CoMP</i>	If intra-base-station UL CoMP and cell-level intra-frequency MRO are both enabled, event A3 reporting in UL CoMP is affected. As a result, the evaluation on UEs in overlapping coverage areas in UL CoMP is affected.
Low-frequency TDD	Intra-base-station DL CoMP	INTRA_GNB_DL_JT_SW option of the NRDUCellAIgoSwitch.CompSwitch parameter	<i>CoMP</i>	If intra-base-station DL CoMP and cell-level intra-frequency MRO are both enabled, event A3 reporting in DL CoMP is affected. As a result, the evaluation on UEs in overlapping coverage areas in DL CoMP is affected.
Low-frequency TDD	Co-MM	INTRA_GNB_SU_COMM_SW option of the NRDUCellCollabServ.MultiCellMimoSwitch parameter	<i>Co-MM (Low-Frequency TDD)</i>	If Co-MM and cell-level intra-frequency MRO are both enabled, event A3 reporting in Co-MM is affected. As a result, the evaluation on UEs in overlapping coverage areas in Co-MM is affected.
Low-frequency TDD	Coordinated interference management	INTRA_GNB_DL_CS_CBF_SW or INTER_GNB_DL_CS_CBF_SW option of the NRDUCellAIgoSwitch.CompSwitch parameter	<i>Coordinated Interference Management (Low-Frequency TDD)</i>	If coordinated interference management and cell-level intra-frequency MRO are both enabled, event A3 reporting in coordinated interference management is affected. As a result, the evaluation on UEs in overlapping coverage areas in coordinated interference management is affected.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Timing carrier shutdown	TIMING_CARRIER_SHUTDOWN_SW option of the NRDUCellAIgoSwitch.PowerSavingSwitch parameter	<i>Energy Conservation and Emission Reduction</i>	When timing carrier shutdown and cell-level intra-frequency MRO are both enabled, the number of handovers within a period decreases if energy saving lasts for a long time. As a result, MRO may take effect later than expected.
Low-frequency TDD	Low power consumption mode	LOW_PWR_MODE_SW option of the NRDUCellAIgoSwitch.PowerSavingSwitch parameter	<i>Green Site</i>	When low power consumption mode and cell-level intra-frequency MRO are both enabled, the number of handovers within a period decreases if energy saving lasts for a long time. As a result, MRO may take effect later than expected.
Low-frequency TDD	Intelligent carrier shutdown	NRCellRelation.PowerSavingCellFlag	<i>Energy Conservation and Emission Reduction</i>	When intelligent carrier shutdown and cell-level intra-frequency MRO are both enabled, the number of handovers within a period decreases if energy saving lasts for a long time. As a result, MRO may take effect later than expected.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Multi-RF-module coordinated energy saving	ENERGY_EFFICIENCY_BASSED_HO option of the NRCellAlgoSwitch.InterFreqHoSwitch parameter and ENERGY_EFF_BASED_COLLAB_SCH_SW option of the NRCellAlgoSwitch.MultiFreqAlgoSwitch parameter	<i>Energy Conservation and Emission Reduction</i>	When multi-RF-module coordinated energy saving and cell-level intra-frequency MRO are both enabled, event A3 reporting in multi-RF-module coordinated energy saving is affected. As a result, the downlink experience evaluation result of the serving cell may be different from the expected result, which may cause the average downlink UE throughput to decrease.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Fast carrier shutdown	Capacity-layer cells: INTRA_GNB_MULTICARRIER_SHUTDOWN_SWITCH option of the NRDUCellAlgoSwitch.PowerSavingsSwitch parameter (for intra-base-station multi-carrier scenarios)/ INTER_GNB_MULTICARRIER_SHUTDOWN_SWITCH option of the NRDUCellAlgoSwitch.PowerSavingsSwitch parameter (for inter-base-station multi-carrier scenarios) FAST_CARRIER_SHUTDOWN_SWITCH option of	<i>Energy Conservation and Emission Reduction</i>	When fast carrier shutdown and cell-level intra-frequency MRO are both enabled, the number of handovers within a period decreases if energy saving lasts for a long time. As a result, MRO may take effect later than expected.

RAT	Function Name	Function Switch	Reference	Description
		the NRDUCellAlgoSwitch.PowerSavingExtSwitch parameter Basic-layer cells: FAST_CARRIER_SHUT_SERV_ASSUR_S W option of the NRDUCellAlgoSwitch.PowerSavingExtSwitch parameter		
Low-frequency TDD	LTE and NR intelligent carrier shutdown	LNR_SMART_CARRIER_SHUTDOWN_SW option of the NRDUCellAlgoSwitch.PowerSavingSwitch parameter	<i>Multi-RAT Coordinated Energy Saving</i>	When LTE and NR intelligent carrier shutdown and cell-level intra-frequency MRO are both enabled, the number of handovers within a period decreases if energy saving lasts for a long time. As a result, MRO may take effect later than expected.
Low-frequency TDD	RF module deep dormancy	RRU.DORMANCYSW set to ON or ON_DEEP_SUPER	<i>Multi-RAT Coordinated Energy Saving</i>	When RF module deep dormancy and cell-level intra-frequency MRO are both enabled, the number of handovers within a period decreases if energy saving lasts for a long time. As a result, MRO may take effect later than expected.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	RF module fast deep dormancy	FAST_DEEP_DORM_SW option of the RRUEXTINFO.RRUDOROPTFEATS parameter	<i>Multi-RAT Coordinated Energy Saving</i>	When RF module fast deep dormancy and cell-level intra-frequency MRO are both enabled, the number of handovers within a period decreases if energy saving lasts for a long time. As a result, MRO may take effect later than expected.
Low-frequency TDD	RF module super dormancy	RRU.DORMANCYSW set to ON_SUPER or ON_DEEP_SUPER	<i>Multi-RAT Coordinated Energy Saving</i>	When RF module super dormancy and cell-level intra-frequency MRO are both enabled, the number of handovers within a period decreases if energy saving lasts for a long time. As a result, MRO may take effect later than expected.
Low-frequency TDD	Coverage-based intra-frequency handover	None	<i>SA Mobility Management in Connected Mode</i>	When coverage-based intra-frequency handover and cell-level intra-frequency MRO are both enabled, CIO adjustment by MRO affects the number of intra-frequency handovers and the RRC connection reestablishment rate.
Low-frequency TDD	Spectral efficiency prediction for downlink experience evaluation	DL_EXP_EV AL_SPCT_EFF_PRED_SW option of the NRCCellAlgoSwitch . MultifreqAlgoSwitch parameter	<i>Multi-Frequency Smart Selection</i>	When spectral efficiency prediction for downlink experience evaluation and cell-level intra-frequency MRO are both enabled, event A3 reporting in spectral efficiency prediction for downlink experience evaluation is affected. As a result, the downlink experience evaluation result of the serving cell may be different from the expected result, which may cause the average downlink UE throughput to decrease.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Downlink spectral efficiency prediction for multi-frequency beam coordination	MF_BEAM_CO_DL_SPECT_EFF_PRED_SW option of the NRCellAlgorithmSwitch.MultiFreqAlgorithmSwitch parameter	<i>Multi-Frequency Smart Selection</i>	When downlink spectral efficiency prediction for multi-frequency beam coordination and cell-level intra-frequency MRO are both enabled, event A3 reporting in downlink spectral efficiency prediction for multi-frequency beam coordination is affected and the downlink experience evaluation result of the serving cell may be different from the expected result. As a result, the average downlink UE throughput may decrease.
Low-frequency TDD	Use of virtual grid models in inter-frequency handovers	NRCellAlgorithmSwitch.AiInterFreqHoOptSwitch	<i>Virtual Grid-based Multi-Frequency Coordination</i>	When use of virtual grid models in inter-frequency handovers and cell-level intra-frequency MRO are both enabled, CIO adjustment by MRO affects the occasion for intra-frequency handovers to trigger event A3. As a result, the number of inter-frequency handovers may be affected.
Low-frequency TDD	Control mode	NRCellRelation.CtrlMode	<i>ANR</i>	If the NRCellRelation.CtrlMode parameter is set to MANUAL_MODE for a neighboring cell, cell-level intra-frequency MRO does not take effect for the neighboring cell.

4.3.3 Hardware

The hardware mentioned in this section must support NR and the configured work modes must include NR.

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910.

DBS3900 LampSite and DBS5900 LampSite. DBS3900 LampSite must be configured with the BBU3910.

Boards

All NR-capable main control boards and baseband processing units support this function. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

RF Modules

All NR-capable RF modules that work in low frequency bands support this function. For details, see the technical specifications of RF modules in *3900 & 5900 Series Base Station Product Documentation*.

4.3.4 Networking

- The Xn link between gNodeBs must be normal, which can be queried by running the **DSP GNBCUXNINTERFACE** command. If there is no Xn link between gNodeBs or the Xn link is faulty, the gNodeB cannot correctly measure the number of premature handovers, delayed handovers, and handovers to a wrong cell because Xn-based Failure Indication messages cannot be properly transmitted. As a result, the accuracy of the handover parameter optimization cannot be ensured.
- Intra-frequency neighboring cells have bidirectional neighbor relationships with each other, which can be queried by running the **LST NRCELLRELATION** command. Neighbor relationships are required for gNodeBs to send Failure Indication and Handover Report messages through Xn links. For unidirectional neighboring cells, gNodeBs may fail to measure some abnormal handovers.

4.3.5 Others

- UEs must comply with 3GPP Release 16 or later.
- Base stations from other vendors are not supported.

4.4 Operation and Maintenance

4.4.1 Data Configuration

4.4.1.1 Data Preparation

[Table 4-5](#) describes the parameters used for function activation.

Table 4-5 Parameters used for activation

Parameter Name	Parameter ID	Option	Setting Notes
Scenario Identification Switch	gNBMrro.ScenarioldentifySwitch	MRO_SCENARIO_IDENT_SW	To enable cell-level MRO, select this option.
Mode Switch	NRCellMrro.ModeSwitch	NR_INTRA_RAT_MRO_CTRL_MODE_SW	Set this option based on the network plan. - To enable the gNodeB to automatically execute the optimization advice, deselect this option. - To enable the optimization advice to be manually confirmed by maintenance personnel before being executed, select this option.
MRO Switch	NRCellMrro.MroSwitch	INTRA_FREQ_MRO_SW	To enable cell-level intra-frequency MRO, select the MRO_SCENARIO_IDENT_SW option of the gNBMrro.ScenarioldentifySwitch parameter first and then select this option.
Optimization Period	gNBMrro.OptPeriod	None	Use the default value.
Statistic Number Threshold	gNBMrro.StatNumThld	None	Use the default value.
Abnormal Handover Ratio Threshold	gNBMrro.AbnHoRatioThld	None	Use the default value.
Too Early and Wrong Handover Ratio Threshold	gNBMrro.TooEarlyHoWrongHoRatioThld	None	Use the default value.
Too Late Handover Ratio Threshold	gNBMrro.TooLateHoRatioThld	None	Use the default value.

4.4.1.2 Using MML Commands

Before using MML commands, refer to [4.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency,

and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

```
//Turning on the MRO scenario identification switch
MOD GNBMMRO: ScenarioIdentifySwitch=MRO_SCENARIO_IDENT_SW-1;
//Optional, required only when the MRO optimization mode needs to be changed to the controlled mode
Turning on the MRO controlled mode switch
MOD NRCELLMRO: NrCellId=0, ModeSwitch=NR_INTRA_RAT_MRO_CTRL_MODE_SW-1;
//Enabling intra-frequency MRO
MOD NRCELLMRO: NrCellId=0, MroSwitch=INTRA_FREQ_MRO_SW-1;
//Optional, required only when cell-level intra-frequency MRO parameters need to be modified
Configuring cell-level intra-frequency MRO parameters
MOD GNBMMRO: OptPeriod=1440, StatNumThld=1000, TooEarlyHoWrongHoRatioThld=70,
TooLateHoRatioThld=70, AbnHoRatioThld=10;
```

Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

```
//Turning off the MRO controlled mode switch
MOD NRCELLMRO: NrCellId=0, ModeSwitch=NR_INTRA_RAT_MRO_CTRL_MODE_SW-0;
//Turning off the intra-frequency MRO switch
MOD NRCELLMRO: NrCellId=0, MroSwitch=INTRA_FREQ_MRO_SW-0;
//Turning off the MRO scenario identification switch
MOD GNBMMRO: ScenarioIdentifySwitch=MRO_SCENARIO_IDENT_SW-0;
```

4.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

4.4.2 Activation Verification

You can query MRO optimization advice or SON logs to verify whether the cell-level intra-frequency MRO has been activated.

NOTE

- If the gNodeB determines not to perform MRO based on the statistics about abnormal handovers, you can only check MRO switch settings and event description in SON logs.
- The values of MRO-related counters in event description of SON logs are the accumulated values within all MRO optimization periods during which handover parameter optimization is triggered.

Free Mode

You can only query SON logs for verification.

- Step 1** On the MAE-Access, choose **SON > SON Log > Synchronize NE Log**.
- Step 2** On the displayed **Synchronize NE Log** tab page, click **Manual Synchronization**. In the displayed dialog box, select NEs and click **OK**.
- Step 3** After the synchronization is completed, click **Query SON Log**. In the **Query SON Log** window, click **Query**. In the displayed list, select **NR MRO Log** from the **Log Category** drop-down list and click **OK**.

The cell-level intra-frequency MRO has been activated if either of the following conditions is met:

- In the **Query SON Log** window, there is a log whose **Event Name** is **Set MRO Switch**, and the MRO switch is set to **ON** in **Event Description**.
- In the **Query SON Log** window, there is a log whose **Event Name** is **Adjust IntraFreq Mobility Parameters**.

----End

Controlled Mode

You can query MRO optimization advice or SON logs to verify the performance.

Querying MRO Optimization Advice

- Step 1** Log in to MAE-Access and choose **SON > MRO > NR MRO**.
- Step 2** On the displayed **NR MRO** page, choose **Optimization Advice > Intra-RAT Handover Optimization Advice**. On the displayed **Intra-RAT Handover Optimization Advice** page, click **Select NE**, select an NE, and click **OK**.

If optimization advice is displayed on the **Intra-RAT Handover Optimization Advice** page, the cell-level intra-frequency MRO has been successfully activated.

----End

Querying SON Logs

- Step 1** On the MAE-Access, choose **SON > SON Log > Synchronize NE Log**.
- Step 2** On the displayed **Synchronize NE Log** tab page, click **Manual Synchronization**. In the displayed dialog box, select NEs and click **OK**.
- Step 3** After the synchronization is completed, click **Query SON Log**. In the **Query SON Log** window, click **Query**. In the displayed list, select **NR MRO Log** from the **Log Category** drop-down list and click **OK**.

The cell-level intra-frequency MRO has been activated if either of the following conditions is met:

- In the **Query SON Log** window, there is a log whose **Event Name** is **Set MRO Switch**, and the MRO switch is set to **ON** in **Event Description**.

- In the **Query SON Log** window, there is a log whose **Event Name** is **Adjust IntraFreq Mobility Parameters** or **Reject Parameters Modify Request**.

----End

4.4.3 Network Monitoring

After this function is enabled, monitor the decrease in the number of abnormal handovers between two cells using the counters in [4.2.1 Benefits](#). In addition, the counters in [Table 4-6](#) can be used to monitor the number of cell-level abnormal handovers.

Table 4-6 MRO-related performance counters

Counter ID	Counter Name
1911837938	N.HO.IntraRAT.HoTooEarly
1911837939	N.HO.IntraRAT.HoToWrgCell
1911837940	N.HO.IntraRAT.HoTooLate
1911837941	N.RLF.CoverageAbnormal

5 UE-Level MRO

5.1 Principles

With UE-level MRO, the gNodeB can automatically identify ping-pong handovers during intra-frequency handovers, collect statistics on abnormal handovers, and automatically optimize UE-level handover parameters based on the statistics and cell-level handover parameters.

 NOTE

Ping-pong handovers described in this document are handovers in a broad sense, including reestablishments.

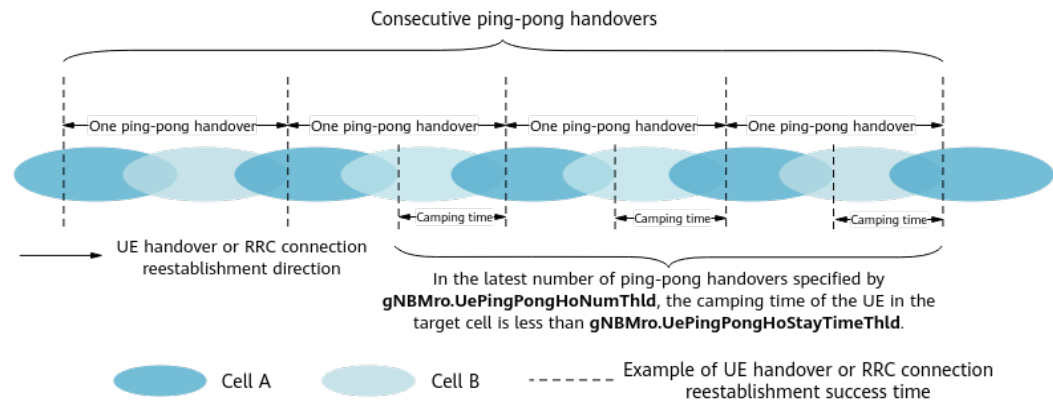
5.1.1 Scenario Identification

A ping-pong handover occurs when the target cell is too likely to meet the handover conditions or the signal quality is unstable. In this case, a UE is handed over or reestablished between two cells, cell A and cell B, and finally handed over or reestablished to the source cell, cell A. After automatically identifying ping-pong handovers, the gNodeB counts the number of ping-pong handovers using counters including **N.HO.IntraRAT.PingPong**, **N.HO.IntraRAT.PingPong.Consecutive**, and **N.HO.NCell.PingPong**.

The gNodeB determines that a UE is a ping-pong UE and the abnormal handover is a ping-pong handover (see [Figure 5-1](#)) when all of the following conditions are met:

- Number of consecutive ping-pong handovers of a UE $\geq$ **gNBMrO.UePingPongHoNumThld**
- In the latest number of ping-pong handovers specified by **gNBMrO.UePingPongHoNumThld**, the camping time of the UE in the target cell is less than **gNBMrO.UePingPongHoStayTimeThld**.

Figure 5-1 Ping-pong handover scenario



The gNodeB determines the historical camping cells and camping time of the UE based on historical messages of the UE. Only the latest 16 camping cells can be recorded.

- In inter-base-station handover scenarios, the gNodeB checks the *UE History Information* IE in the HANDOVER REQUEST message.
- In inter-base-station reestablishment scenarios, the gNodeB checks the *UE History Information* IE in the RETRIEVE UE CONTEXT RESPONSE message.
- In intra-base-station handover or reestablishment scenarios, the gNodeB checks the UE handover history.

5.1.2 Optimization

After identifying ping-pong handovers, the gNodeB automatically optimizes the UE-level handover parameter CIO based on the cell-level handover parameter CIO (NRCellRelation.*CellIndividualOffset*) if the conditions are met. In addition, the gNodeB performs a penalty mechanism to prevent the UE-level CIO from being adjusted to an excessively small value. This function is controlled by the UE_MRO_SWITCH option of the NRCellMro.*MroSwitch* parameter.

Parameter Optimization

After identifying a ping-pong handover, the gNodeB automatically optimizes the UE-level CIO. If the UE_MRO_SWITCH option of the NRCellMro.*MroSwitch* parameter is selected, the gNodeB determines whether the UE's method to move from cell B to the source cell, cell A, in the latest ping-pong handover is a handover or a reestablishment caused by a handover.

- If yes, CIO optimization is planned for a ping-pong UE only in the following scenarios:
 - If the number of consecutive ping-pong handovers of the ping-pong UE is equal to N and the single camping time in the target cell during the N ping-pong handovers is less than T , the gNodeB plans to decrease the CIO value of the UE by one step based on the cell-level CIO value.
 - If the number of consecutive ping-pong handovers is greater than N and the single camping time in the target cell during the latest N ping-pong handovers is less than T , perform the following operation:

- If the camping time in the target cell during the latest ($N+1$)th ping-pong handover is greater than or equal to T , the gNodeB plans to decrease the CIO value of the UE by one step based on the cell-level CIO value.
- If the camping time in the target cell during the latest ($N+1$)th ping-pong handover is less than T , the gNodeB plans to decrease the CIO value of the UE by two steps based on the cell-level CIO value.
- If no, CIO optimization is not performed for the ping-pong UE.

where

- N : **gNBMrro.UePingPongHoNumThld**
- T : **gNBMrro.UePingPongHoStayTimeThld**
- Cell-level CIO: **NRCellRelation.CellIndividualOffset**

During UE-level CIO optimization, the gNodeB will evaluate whether the UE-level CIO value adjusted based on the optimization plan falls into the value range of the CIO parameter **NRCellRelation.CellIndividualOffset** and then determine whether to perform the optimization. Table 5-1 describes the gNodeB's optimization policies for the UE-level CIO.

Table 5-1 CIO optimization policies

Optimization Plan	Whether the Value After Optimization Is Within the Value Range Specified by the CIO Parameter	Optimization Result
Decrease by one step	Yes	Decrease by one step
	No	No optimization
Decrease by two steps	Yes	Decrease by two steps
	No, and the value is within the value range specified by the CIO parameter after being decreased by one step.	Decrease by one step
	No, and the value is beyond the value range specified by the CIO parameter after being decreased by one step.	No optimization

Optimization Penalty

When optimizing the CIO value of a ping-pong UE, the gNodeB uses a penalty mechanism to prevent the CIO value from being too small.

When the **MRO_SCENARIO_IDENT_SW** option of the **gNBMrro.ScenarioidentifySwitch** parameter is selected, a cell-level MRO optimization period specified by **gNBMrro.OptPeriod** starts. The gNodeB does not perform UE-level MRO in the next optimization period if all the following conditions are met in the current optimization period:

- Number of intra-frequency handovers $\geq$ **gNBmro.StatNumThld**
- Proportion of RLF-induced abnormal handovers $>$ **gNBmro.AbnHoRatioThld**
- Proportion of delayed handovers to RLF-induced abnormal handovers $>$ **gNBmro.TooLateHoRatioThld**

For details about the number of intra-frequency handovers, proportion of RLF-induced abnormal handovers, and number of RLF-induced abnormal handovers, see [Parameter Optimization](#).

NOTE

The current optimization period refers to cumulative optimization periods during which the number of intra-frequency handovers is greater than or equal to the value of **gNBmro.StatNumThld**.

If UE-level MRO and cell-level intra-frequency MRO are both enabled, a cell-level MRO optimization period specified by **gNBmro.OptPeriod** starts when the **INTRA_FREQ_MRO_SW** option of the **NRCelMro.MroSwitch** parameter is selected. The gNodeB clears the statistics before cell-level intra-frequency MRO is enabled and restarts the statistics measurement.

5.2 Network Analysis

5.2.1 Benefits

UE-level MRO reduces the number of ping-pong handovers and air interface messages, and decreases the values of the **N.HO.IntraRAT.PingPong**, **N.HO.IntraRAT.PingPong.Consecutive**, and **N.HO.NCell.PingPong** counters, reducing network planning, optimization, and O&M costs and improving network optimization efficiency.

It is recommended that UE-level MRO be enabled when the proportion of ping-pong handovers to the total number of handovers is greater than 10% and the proportion of consecutive ping-pong handovers to the total number of handovers is high. If the preceding conditions are not met, the gains provided by this function are not obvious.

where

- The number of ping-pong handovers is measured by the **N.HO.IntraRAT.PingPong** counter.
- The number of consecutive ping-pong handovers is measured by the **N.HO.IntraRAT.PingPong.Consecutive** counter.
- Total number of handovers = **N.HO.Xn.IntergNB.ExecSucIn** + **N.HO.Ng.IntergNB.ExecSucIn** + **N.HO.IntragNB.ExecSucIn**

5.2.2 Impacts

The impacts between this function and other functions are described in [5.3.2.3 Function Impacts](#).

During UE-level MRO against ping-pong handovers, if the CIO is adjusted to a value too small, handovers may not be triggered even if the signal quality in the

neighboring cell is much higher than that in the serving cell. As a result, the probability of delayed handovers increases, the channel quality of some UEs in the serving cell deteriorates, and therefore the throughput decreases.

Throughput-related KPIs: **User Downlink Average Throughput (DU)**, **User Uplink Average Throughput (DU)**, **Cell Downlink Average Throughput (DU)**, and **Cell Uplink Average Throughput (DU)**

5.3 Requirements

5.3.1 Licenses

Feature ID	Feature Name	Model	Sales Unit
FOFD-070320	Mobility Robustness Optimization	NR0S00MRPT00	per Cell

NOTE

The insufficiency of certain feature licenses affects cell activation. For details, see the **Is Cell Activation Affected by License Insufficiency** column in *License Control Item Lists*.

5.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

5.3.2.1 Prerequisite Functions

RAT	Function Name	Function Switch	Reference
Low-frequency TDD	Cell-level MRO scenario identification	MRO_SCENARIO_IDENT_SW option of the gNBMrro.ScenarioIdentify-Switch parameter	<i>MRO</i>

5.3.2.2 Mutually Exclusive Functions

None

5.3.2.3 Function Impacts

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Intra-base-station UL CoMP	INTRA_GNB_UL_COMP_SW option of the NRDUCellAIgoSwitch.CompSwitch parameter	<i>CoMP</i>	When intra-base-station UL CoMP and UE-level MRO are both enabled and UL CoMP takes effect for ping-pong UEs, event A3 reporting in UL CoMP is affected. As a result, the evaluation on UEs in overlapping coverage areas in UL CoMP is affected.
Low-frequency TDD	Intra-base-station DL CoMP	INTRA_GNB_DL_JT_SW option of the NRDUCellAIgoSwitch.CompSwitch parameter	<i>CoMP</i>	When intra-base-station DL CoMP and UE-level MRO are both enabled and DL CoMP takes effect for ping-pong UEs, event A3 reporting in DL CoMP is affected. As a result, the evaluation on UEs in overlapping coverage areas in DL CoMP is affected.
Low-frequency TDD	Co-MM	INTRA_GNB_SU_COMM_SW option of the NRDUCellCollabServ.MultiCellMimoSwitch parameter	<i>Co-MM (Low-Frequency TDD)</i>	If both Co-MM and UE-level MRO are enabled and Co-MM takes effect for ping-pong UEs, event A3 reporting in Co-MM is affected. As a result, the evaluation on UEs in overlapping coverage areas in Co-MM is affected.
Low-frequency TDD	Coordinated interference management	INTRA_GNB_DL_CS_CBF_SW or INTER_GNB_DL_CS_CBF_SW option of the NRDUCellAIgoSwitch.CompSwitch parameter	<i>Coordinated Interference Management (Low-Frequency TDD)</i>	If coordinated interference management and UE-level MRO are both enabled and coordinated interference management takes effect for ping-pong UEs, event A3 reporting in coordinated interference management is affected. As a result, the evaluation on UEs in overlapping coverage areas in coordinated interference management is affected.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Timing carrier shutdown	TIMING_CARRIER_SHUTDOWN_SW option of the NRDUCellAIgoSwitch.PowerSavingSwitch parameter	<i>Energy Conservation and Emission Reduction</i>	When timing carrier shutdown and UE-level MRO are both enabled, the number of handovers within a period decreases if energy saving lasts for a long time. As a result, the penalty mechanism accuracy of UE-level MRO may be affected.
Low-frequency TDD	Low power consumption mode	LOW_PWR_MODE_SW option of the NRDUCellAIgoSwitch.PowerSavingSwitch parameter	<i>Green Site</i>	When both low power consumption mode and UE-level MRO are enabled, the number of handovers within a period decreases if energy saving lasts for a long time. As a result, the penalty mechanism accuracy of UE-level MRO may be affected.
Low-frequency TDD	Intelligent carrier shutdown	NRCCellRelation.PowerSavingCellFlag	<i>Energy Conservation and Emission Reduction</i>	When intelligent carrier shutdown and UE-level MRO are both enabled, the number of handovers within a period decreases if energy saving lasts for a long time. As a result, the penalty mechanism accuracy of UE-level MRO may be affected.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Multi-RF-module coordinated energy saving	ENERGY_EFFICIENCY_BASED_HO option of the NRCellAlgoSwitch.InterFreqHoSwitch parameter and ENERGY_EFF_BASED_COLLAB_SCH_SW option of the NRCellAlgoSwitch.MultiFreqAlgoSwitch parameter	<i>Energy Conservation and Emission Reduction</i>	When multi-RF-module coordinated energy saving and UE-level MRO are both enabled, event A3 reporting will be affected if multi-RF-module coordinated energy saving takes effect for ping-pong UEs. As a result, the downlink experience evaluation result of the source cells in ping-pong handovers may be different from the expected result, which may cause the average downlink UE throughput to decrease.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Fast carrier shutdown	Capacity-layer cells: • INTRA_GNB_MULTICARRIER_SHUTDOWN_SW option of the NRDUCellAlgoSwitch.PowerSavingsSwitch parameter (for intra-base-station multi-carrier scenarios)/ INTER_GNB_MULTICARRIER_SHUTDOWN_SW option of the NRDUCellAlgoSwitch.PowerSavingsSwitch parameter (for inter-base-station multi-carrier scenarios) • FAST_CARRIER_SHUTDOWN_SW option of	<i>Energy Conservation and Emission Reduction</i>	When fast carrier shutdown and UE-level MRO are both enabled, the number of handovers within a period decreases if energy saving lasts for a long time. As a result, the penalty mechanism accuracy of UE-level MRO may be affected.

RAT	Function Name	Function Switch	Reference	Description
		the NRDUCellAlgoSwitch.PowerSavingExtSwitch parameter Basic-layer cells: FAST_CARRIER_SHUT_DOWN_ASSURANCE option of the NRDUCellAlgoSwitch.PowerSavingExtSwitch parameter		
Low-frequency TDD	LTE and NR intelligent carrier shutdown	LNR_SMART_CARRIER_SHUTDOWN_SW option of the NRDUCellAlgoSwitch.PowerSavingSwitch parameter	<i>Multi-RAT Coordinated Energy Saving</i>	When LTE and NR intelligent carrier shutdown and UE-level MRO are both enabled, the number of handovers within a period decreases if energy saving lasts for a long time. As a result, the penalty mechanism accuracy of UE-level MRO may be affected.
Low-frequency TDD	RF module deep dormancy	RRU.DORMANCYSW set to ON or ON_DEEP_SUPER	<i>Multi-RAT Coordinated Energy Saving</i>	When RF module deep dormancy and UE-level MRO are both enabled, the number of handovers within a period decreases if energy saving lasts for a long time. As a result, the penalty mechanism accuracy of the UE-level MRO is affected.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	RF module fast deep dormancy	FAST_DEEP_DORM_SW option of the RRUEXTINFO.RRUDOROPTFEATS W parameter	<i>Multi-RAT Coordinated Energy Saving</i>	When RF module fast deep dormancy and UE-level MRO are both enabled, the number of handovers within a period decreases if energy saving lasts for a long time. As a result, the penalty mechanism accuracy of UE-level MRO may be affected.
Low-frequency TDD	RF module super dormancy	RRU.DORMANCYSW set to ON_SUPER or ON_DEEP_SUPER	<i>Multi-RAT Coordinated Energy Saving</i>	When RF module super dormancy and UE-level MRO are both enabled, the number of handovers within a period decreases if energy saving lasts for a long time. As a result, the penalty mechanism accuracy of the UE-level MRO is affected.
Low-frequency TDD	Coverage-based intra-frequency handover	None	<i>SA Mobility Management in Connected Mode</i>	When coverage-based intra-frequency handover and UE-level MRO are both enabled, CIO adjustment by MRO affects the number of intra-frequency handovers.
Low-frequency TDD	Spectral efficiency prediction for downlink experience evaluation	DL_EXP_EV AL_SPCT_EFF_PRED_SW option of the NRCellAlgoSwitch.MultiFreqAlgoSwitch parameter	<i>Multi-Frequency Smart Selection</i>	When spectral efficiency prediction for downlink experience evaluation and UE-level MRO are both enabled, event A3 reporting in spectral efficiency prediction for downlink experience evaluation is affected for UEs involved in ping-pong handovers and engaged in spectral efficiency prediction for downlink experience evaluation. As a result, the downlink experience evaluation result of the source cells in ping-pong handovers may be different from the expected result, which may cause the average downlink UE throughput to decrease.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Downlink spectral efficiency prediction for multi-frequency beam coordination	MF_BEAM_CO_DL_SPECT_EFF_PRED_SW option of the NRCellAlgoSwitch.MultiFreqAlgoSwitch parameter	<i>Multi-Frequency Smart Selection</i>	When downlink spectral efficiency prediction for multi-frequency beam coordination and UE-level MRO are both enabled, event A3 reporting in downlink spectral efficiency prediction for multi-frequency beam coordination is affected for UEs involved in ping-pong handovers and engaged in downlink spectral efficiency prediction for multi-frequency beam coordination. As a result, the downlink experience evaluation result of the source cells in ping-pong handovers may be different from the expected result, which may cause the average downlink UE throughput to decrease.
Low-frequency TDD	Use of virtual grid models in inter-frequency handovers	NRCellAlgoSwitch.AlnInterFreqHoOptSwitch	<i>Virtual Grid-based Multi-Frequency Coordination</i>	If use of virtual grid models in inter-frequency handovers and UE-level MRO are both enabled and use of virtual grid models in inter-frequency handovers takes effect for ping-pong UEs, CIO adjustment by MRO affects the occasion for intra-frequency handovers to trigger event A3. As a result, the number of inter-frequency handovers may be affected.
Low-frequency TDD	Multiple BWP1s for RedCap	NRDUCellRedCap.RedCapUeMaxAvailableBandwidth set to a value other than 20M	<i>RedCap</i>	UE-level MRO does not take effect on RedCap UEs using BWP1s with NCD-SSBs.

5.3.3 Hardware

The hardware mentioned in this section must support NR and the configured work modes must include NR.

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910.

DBS3900 LampSite and DBS5900 LampSite. DBS3900 LampSite must be configured with the BBU3910.

Boards

All NR-capable main control boards and baseband processing units support this function. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

RF Modules

All NR-capable RF modules that work in low frequency bands support this function. For details, see the technical specifications of RF modules in *3900 & 5900 Series Base Station Product Documentation*.

5.3.4 Networking

Only optimization penalty has the following requirements:

- The Xn link between gNodeBs must be normal, which can be queried by running the **DSP GNBCUXNINTERFACE** command. If there is no Xn link between gNodeBs or the Xn link is faulty, the gNodeB cannot correctly measure the number of premature handovers, delayed handovers, and handovers to a wrong cell because Xn-based Failure Indication messages cannot be properly transmitted. As a result, the accuracy of the penalty mechanism cannot be ensured.
- Intra-frequency neighboring cells have bidirectional neighbor relationships with each other, which can be queried by running the **LST NRCELLRELATION** command. Neighbor relationships are required for gNodeBs to send Failure Indication and Handover Report messages through Xn links. For unidirectional neighboring cells, gNodeBs may fail to measure some abnormal handovers. As a result, the accuracy of the penalty mechanism cannot be ensured.

5.3.5 Others

Only optimization penalty has the following requirements:

- UEs must comply with 3GPP Release 16 or later.
- Base stations from other vendors are not supported.

5.4 Operation and Maintenance

5.4.1 Data Configuration

5.4.1.1 Data Preparation

Table 5-2 describes the parameters used for function activation.

Table 5-2 Parameters used for activation

Parameter Name	Parameter ID	Setting Notes
MRO Switch	NRCeMro.MroSwitch	To enable UE-level MRO, select the UE_MRO_SWITCH option.
UE Ping-Pong HO Number Threshold	gNB Mro.UePingPongHoNumThld	Use the default value.
UE Ping-Pong HO Stay Time Threshold	gNB Mro.UePingPongHoStayTimeThld	Use the default value.
Optimization Period	gNB Mro.OptPeriod	Use the default value.
Statistic Number Threshold	gNB Mro.StatNumThld	Use the default value.
Abnormal Handover Ratio Threshold	gNB Mro.AbnHoRatioThld	Use the default value.
Too Late Handover Ratio Threshold	gNB Mro.TooLateHoRatioThld	Use the default value.

5.4.1.2 Using MML Commands

Before using MML commands, refer to [5.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

```
//Turning on the UE-level MRO switch
MOD NRCeLMRO: NrCellId=0, MroSwitch=UE_MRO_SWITCH-1;
//(Optional, required only when UE-level MRO parameters need to be modified) Configuring UE-level MRO parameters
MOD GNB MRO: OptPeriod=1440, StatNumThld=1000, TooLateHoRatioThld=70, AbnHoRatioThld=10, UePingPongHoNumThld=2, UePingPongHoStayTimeThld=2;
```

Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

```
//Turning off the UE-level MRO switch  
MOD NRCELLMRO: NrCellId=0, MroSwitch=UE_MRO_SWITCH-0;
```

5.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

5.4.2 Activation Verification

You can observe counters to check whether UE-level MRO has taken effect.

If the **N.HO.IntraRAT.UeMro.CIO** counter produces a non-zero value, UE-level MRO has taken effect on the network provided that ping-pong UEs exist on the network.

5.4.3 Network Monitoring

After this function is enabled, observe counters in [5.2.1 Benefits](#) to monitor the decrease in the number of ping-pong handovers. In addition, the **N.HO.IntraRAT.UeMro.CIO** counter can be used to observe the changes in the number of UE-level CIO adjustments.

6 Parameters

The following hyperlinked EXCEL files of parameter reference match the software version with which this document is released.

- Node Parameter Reference: contains device and transport parameters.
- Node Used Reserved Parameter List: contains the reserved parameters that are in use and those that have been disused.
- gNodeBFunction Parameter Reference: contains all parameters related to radio access functions, including air interface management, access control, mobility control, and radio resource management.
- 3900 & 5900 Series Base Station gNodeBFunction Used Reserved Parameter List: contains the reserved parameters that are in use and the reserved parameters that fall into disuse in the current R version.

NOTE

You can find the EXCEL files of parameter reference and used reserved parameter list for the software version used on the live network from the "Man-Machine Interface Reference" node in the *3900 & 5900 Series Base Station Product Documentation* delivered with that version.

FAQ 1: How do I find the parameters related to a certain feature from parameter reference?

- Step 1** Open the EXCEL file of parameter reference.
- Step 2** On the **Parameter List** sheet, filter the **Feature ID** column. Click **Text Filters** and choose **Contains**. Enter the feature ID, for example, FBFD-010011.
- Step 3** Click **OK**. All parameters related to the feature are displayed.

----End

FAQ 2: How do I find the information about a certain reserved parameter from the used reserved parameter list?

- Step 1** Open the EXCEL file of the used reserved parameter list.

Step 2 On the **Used Reserved Parameter List** sheet, use the **MO**, **Parameter ID**, and **BIT** columns to locate the reserved parameter. View its information, including the meaning, values, and impacts.

----End

7 Counters

The following hyperlinked EXCEL files of performance counter reference match the software version with which this document is released.

- Node Performance Counter Summary: contains device and transport counters.
- gNodeBFunction Performance Counter Summary: contains all counters related to radio access functions, including air interface management, access control, mobility control, and radio resource management.
- gNodeBFunction Used Reserved Counter List: contains the reserved counters that are in use and those that have been disused.

NOTE

You can find the EXCEL files of performance counter reference and used reserved counter list for the software version used on the live network from the "Man-Machine Interface Reference" node in the *3900 & 5900 Series Base Station Product Documentation* delivered with that version.

FAQ: How do I find the counters related to a certain feature from performance counter reference?

- Step 1** Open the EXCEL file of performance counter reference.
- Step 2** On the **Counter Summary(En)** sheet, filter the **Feature ID** column. Click **Text Filters** and choose **Contains**. Enter the feature ID, for example, FBFD-010011.
- Step 3** Click **OK**. All counters related to the feature are displayed.

----End

8 Glossary

For the acronyms, abbreviations, terms, and definitions, see *Glossary*.

9 Reference Documents

- *CoMP*
- *Co-MM (Low-Frequency TDD)*
- *Coordinated Interference Management (Low-Frequency TDD)*
- *Green Site*
- *Energy Conservation and Emission Reduction*
- *Multi-RAT Coordinated Energy Saving*
- *SA Mobility Management in Connected Mode*
- *Multi-Frequency Smart Selection*
- *Virtual Grid-based Multi-Frequency Coordination*
- *ANR*
- *RedCap*
- *License Control Item Lists*
- *High Speed Mobility*